



SC-Unext: A Lightweight Image Segmentation Model with Cellular Mechanism for Breast Ultrasound Tumor Diagnosis

Fenglin Cai¹ · Jiaying Wen² · Fangzhou He¹ · Yulong Xia² · Weijun Xu² · Yong Zhang² · Li Jiang² · Jie Li¹

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Abstract

Automatic breast ultrasound image segmentation plays an important role in medical image processing. However, current methods for breast ultrasound segmentation suffer from high computational complexity and large model parameters, particularly when dealing with complex images. In this paper, we take the Unext network as a basis and utilize its encoder-decoder features. And taking inspiration from the mechanisms of cellular apoptosis and division, we design apoptosis and division algorithms to improve model performance. We propose a novel segmentation model which integrates the division and apoptosis algorithms and introduces spatial and channel convolution blocks into the model. Our proposed model not only improves the segmentation performance of breast ultrasound tumors, but also reduces the model parameters and computational resource consumption time. The model was evaluated on the breast ultrasound image dataset and our collected dataset. The experiments show that the SC-Unext model achieved Dice scores of 75.29% and accuracy of 97.09% on the BUSI dataset, and on the collected dataset, it reached Dice scores of 90.62% and accuracy of 98.37%. Meanwhile, we conducted a comparison of the model's inference speed on CPUs to verify its efficiency in resource-constrained environments. The results indicated that the SC-Unext model achieved an inference speed of 92.72 ms per instance on devices equipped only with CPUs. The model's number of parameters and computational resource consumption are 1.46M and 2.13 GFlops, respectively, which are lower compared to other network models. Due to its lightweight nature, the model holds significant value for various practical applications in the medical field.

Keywords Breast ultrasound · Deep learning · Tumor segmentation · Lightweight

Introduction

Medical image segmentation [1–3] is an important area in the field of medical imaging, playing a crucial role in various aspects, including disease diagnosis, treatment

planning, and patient monitoring [4]. Accurately delineating anatomical structures or pathological regions allows medical professionals to gain a deeper understanding of image content and make precise diagnoses and decisions. In recent years, with the rapid expansion of medical imaging data and the advancements in computer vision, medical

Fenglin Cai and Jiaying Wen contributed equally to this work.

✉ Li Jiang
drjiangli2019@163.com

✉ Jie Li
jieli@cqust.edu.cn

Fenglin Cai
caiflls@163.com

Jiaying Wen
etoile@stu.cqmu.edu.cn

Fangzhou He
ark125@163.com

Yulong Xia
2010473582@qq.com

Weijun Xu
945685773@qq.com

Yong Zhang
yongcqmu@163.com

¹ Department of Intelligent Technology and Engineering, Chongqing University of Science and Technology, Chongqing 401331, People's Republic of China

² Department of Neurosurgery, The First Affiliated Hospital of Chongqing Medical University, Chongqing 400016, People's Republic of China

image segmentation algorithms have garnered significant attention and research efforts. However, medical image segmentation [5, 6] continues to face multiple challenges and complexities. Firstly, Medical imaging often exhibits complex structures and diverse lesion types, presenting significant challenges to traditional segmentation algorithms [7]. Variations in lesion appearance, such as size, shape, color, and texture, contribute to poor segmentation performance, while low contrast and high noise in imaging modalities like ultrasound or MRI obscure lesion boundaries. Furthermore, medical imaging datasets typically have high noise and non-uniformity, complicating segmentation tasks. Additionally, real-time or near-real-time processing speed is a crucial factor for medical image segmentation tasks.

Breast tumors represent one of the common breast diseases among women [8]. Malignant breast tumors can lead to patient mortality and stand as a major cause of cancer-related deaths globally [9, 10]. Even benign breast masses may elevate the risk of breast cancer [11–13], underscoring the significance of early screening and close monitoring [14, 15]. Breast ultrasound (B-ultrasound) technology ranks among the most commonly employed techniques for early breast mass screening [16]. Early detection and diagnosis [17] of breast cancer yield an overall survival rate as high as 92%. Due to its safety, tolerability, and real-time capabilities, breast ultrasound has gained extensive use in the diagnosis of breast lesions [18, 19]. Accurately segmenting breast lesions from breast ultrasound images stands as a critical step in computer-aided diagnosis [20], facilitating the diagnosis and treatment of breast cancer and ultimately reducing mortality rates. However, currently prevalent image segmentation models often possess a large number of parameters and demand substantial computational resources, frequently requiring considerable computing power to ensure smooth operation. In clinical medical practice, the trend toward miniaturization and portability of devices like ultrasound units is vital [21]. Nevertheless, this trend may lead to ultrasound devices lacking the necessary computational power for automated image segmentation [22, 23].

Hence, this study proposes a lightweight model [24] capable of automatically segmenting breast ultrasound [25] images in resource-constrained scenarios. Building upon the lightweight attributes of the Unext [26] segmentation network model, this research incorporates the SCConv (Spatial and Channel Convolution) block [27] to replace conventional convolution modules. Furthermore, drawing inspiration from the mechanisms of cellular activities in multicellular organisms, algorithms are designed to simulate apoptosis and division, thereby reducing the model's parameter count and enhancing the accuracy and generalization capabilities of the Unext model. This study

validates the improved model using both the BUSI dataset and a private dataset of breast ultrasound images.

Materials and Methods

Dataset

In this study, we performed model validation using the 2018 Breast Ultrasound Scans for Tumor Diagnosis (BUSI) dataset [28], comprising 780 breast ultrasound images from 600 women aged 25 to 75. The images, in PNG format and averaging 500x500 pixels, include tumor masks and original images, categorized into three classes: normal, benign, and malignant (133 normal, 437 benign, 210 malignant). Our primary objective is precise breast tumor segmentation, necessitating the exclusion of non-tumor images. The BUSI dataset encompasses a comprehensive spectrum of breast lesions (benign, malignant, and normal), enabling our model to accurately detect and segment various lesion types. It also facilitates adaptation to the inherent variability in breast ultrasound imaging, providing a thorough understanding of different pathological presentations. Therefore, Many researchers have used the BUSI dataset to validate the effectiveness of their methods [26, 29, 30].

Additionally, BUSI dataset images replicate real-world clinical scenarios, varying in quality, noise, and artifacts, allowing us to assess our model's robustness under challenging imaging conditions. Expert radiologist annotations for lesion boundaries in each BUSI dataset image serve as ground truth for training and validation. The dataset maintains a balanced representation of lesion and non-lesion cases, reducing model bias and ensuring equal proficiency in lesion presence and absence detection. This balance minimizes false positives and negatives, critical in clinical settings.

In addition to the BUSI dataset, this study has collected breast ultrasound data from patients as its primary research subject. The study's inclusion criteria encompass the following: (1) Complete clinical data; (2) Confirmation of breast cancer or benign breast tumors through pathological examination. Meanwhile, the exclusion criteria entail the following: (1) Prior chemotherapy, radiation therapy, or surgical treatment before hospitalization; (2) Severe comorbidities affecting other organs; (3) Metabolic disorders, blood system diseases, or infectious diseases; (4) Other malignancies; (5) Severe obesity.

The breast ultrasound dataset collected for this study included 915 images from 305 patients between 2022 and 2023. Among these patients, 167 had breast cancer with an average age of (29.15 ± 2.83) years, and 138 had benign breast tumors with an average age of (28.68 ± 2.27) years.

All patients were female. To annotate breast lesion regions in the ultrasound images, we followed established medical image segmentation annotation procedures. We invited two professional ultrasound doctors with over 20 years of experience to use the Labelme tool to annotate the lesion regions in each breast ultrasound image. In cases of disagreement during the annotation process, a chief ultrasound doctor with over 25 years of experience confirmed the annotated data.

Methodology

Currently, most breast ultrasound tumor segmentation models focus on improving accuracy, resulting in an increase in model parameters and suboptimal inference speed. Particularly in the context of practical medical image applications, where many devices lack GPU capabilities, model lightweighting becomes crucial. In comparison to Unet [31], the Unext network still adheres to the encoder-decoder structure akin to Unet. However, due to the inherent simplicity of MLP (Multi-Layer Perceptron) as opposed to Convolution and Transformer methods, it significantly reduces

computational complexity, rendering the model more lightweight and suitable for practical applications.

In this study, we propose enhancements to the Unext network by introducing the Apoptosis and Division algorithm, simulating cell apoptosis and division processes to optimize the network model. Additionally, we incorporate the SCConv module to replace standard convolutional modules, thereby improving model performance while achieving lightweighting. The modified network architecture is illustrated in Fig. 1. Notably, in the network model, the channel numbers for C1 to C5 are set at 32, 64, 128, 160, and 256, respectively. In the SCConv Stage, encoders C1, C2, and C3, along with their corresponding decoders, are substituted with SCConv convolution modules.

SCConv Block

The SCConv block consists of two units: the Spatial Reconstruction Unit (SRU) and the Channel Reconstruction Unit (CRU), as illustrated in Fig. 2. SRU utilizes scaling factors from group normalization to assess the information content

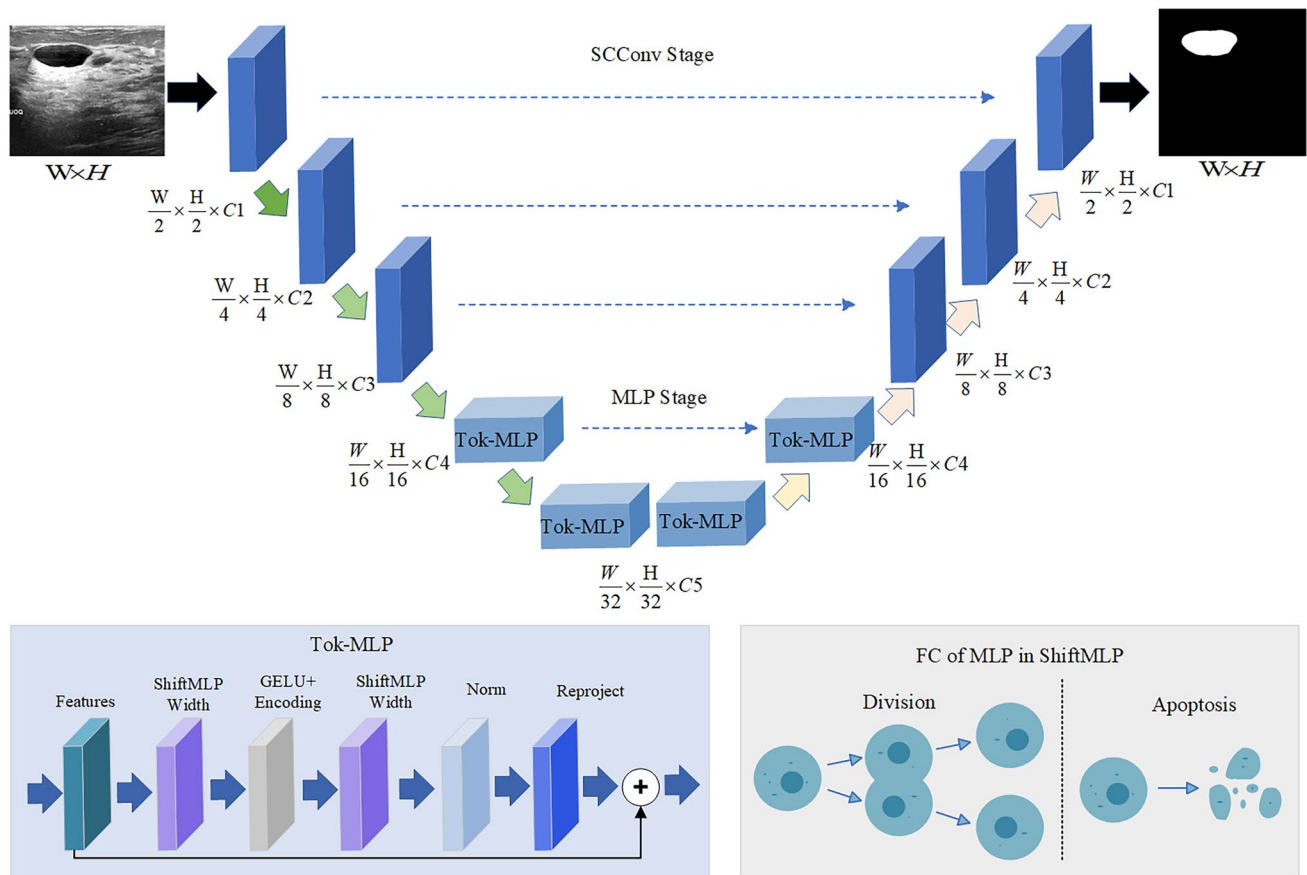


Fig. 1 Overall of the proposed method. C1-C5 are the number of channels, which are 32, 64, 128, 160, and 256 respectively. ShiftMLP is a part of the Tok-MLP module of the MLP Stage. Division and Apoptosis algorithm integrated in ShiftMLP

in different feature maps, categorizing them into informative and non-informative sections. This process enables feature reconstruction, maintaining key information while reducing spatial dimensionality redundancy. CRU utilizes a Split-Transform-and-Fuse strategy. The input feature map is divided into two parts using a predetermined threshold. These parts are then processed through group convolution and point-wise convolution for feature extraction. This approach aims to reduce computational costs while preserving essential features. An adaptive fusion technique is then employed to emphasize information-rich features, thereby reducing feature redundancy in the channel dimension and enhancing feature representation. The SCConv module offers an effective approach to harness spatial and channel redundancies efficiently. It aims to restrict feature redundancy, thereby reducing the number of model parameters and floating-point operations (FLOPs), while enhancing feature representation capability. This leads to improved model performance by reducing redundant features.

Apoptosis and Division

In the MLP Stage, inspired by cellular division and apoptosis, the apoptosis and division algorithms are introduced in the fully connected layers of the ShiftMLP. This approach allows the model to effectively update learned weight parameters, enhancing accuracy and generalization. The apoptosis algorithm, akin to biological cell death, utilizes a preset threshold to filter out redundant neurons, focusing on more significant neuron weight features, thereby reducing network complexity. The implementation process of this algorithm is detailed in Algorithm 1.

Algorithm 1 Apoptosis.

Input: FC bias tensor T , features channel N
Output: Output tensor X

```

1:  $X \leftarrow T$ 
2:  $Id \leftarrow (x \leq -0.03).nonzero(as\_tuple = True)[0]$ 
3:  $d \leftarrow X.index$ 
4:  $p \leftarrow Id.size()[0]/N$ 
5:  $X \leftarrow d/(1.0 - p)$ 
return  $X$ 

```

In Algorithm 1, the input bias tensor x is filtered to identify elements within the bias tensor that are less than or equal

to a threshold value. These elements are set to zero, and then the entire tensor is scaled according to a specified ratio. This process aims to eliminate weight features that have a negative impact, thereby optimizing the network model.

The division algorithm in the SC-Unext model, guided by a preset threshold, selects neural weights with a substantial impact. This significant weight feature undergoes a division operation, being reassigned to neurons with lesser influence. This process serves to refine the model's structure, as delineated in Algorithm 2.

Algorithm 2 Division.

Input: Apoptosis tensor T , features channel N , bias thresholds T
Output: Output tensor X

```

1:  $X \leftarrow T$ 
2:  $q \leftarrow (x \leq -t).nonzero(as\_tuple = True)[0]$ 
3: if  $q.size()[0] \geq 2$  then
4:    $id \leftarrow \text{argmax}(T)$ 
5:    $a \leftarrow \text{tensor}([id.data])$ 
6:    $g \leftarrow \text{index\_select}(X, 1, a)$ 
7:   while  $q.size()[0] \geq 2$  do
8:      $t \leftarrow q.size()[0]$ 
9:      $d \leftarrow \text{randint}(0)$ 
10:     $q \leftarrow \text{torch.cat}([q[0:d], q[d+1:]], dim = 0)$ 
11:   end while
12:    $z1 \leftarrow \text{torch.zeros}(16, N)$ 
13:    $z2 \leftarrow z1.index\_fill(1, q, 1)$ 
14:    $X \leftarrow z2 * g + X$ 
15: end if
return  $X$ 

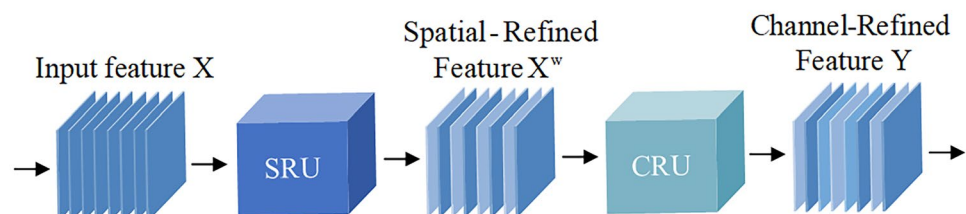
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In Algorithm 2, steps 1 to 2 filter the input feature tensor to obtain $id1$, which is used to determine the feature data that needs to be updated. Steps 3 to 11 involve the selection and modification of feature weights within the bias tensor. Steps 12 to 14 are involved in computing the feature tensor. By selectively modifying the feature tensor after apoptosis operations, specific conditions for feature data are randomly chosen and updated, resulting in optimized features through the division algorithm, ultimately enhancing the model's performance.

Metrics for Ultrasound Breast Lesions Segmentation

In the context of this study, model development is implemented utilizing the PyTorch framework, with training conducted using a combination of binary cross-entropy loss and Dice loss functions. Specifically, the binary cross-entropy loss function serves to quantify the dissimilarity between the model's predictions

Fig. 2 Architecture of SCConv block



and the ground truth labels, thereby assessing the divergence between predicted and actual values. The mathematical formulation for this loss function is depicted in Eq. (1).

$$Loss_{bce} = -\frac{1}{N} \sum_{i=1}^N y_i * \log(p(y_i)) * \log(1 - p(y_i)) \quad (1)$$

where y is the binary label 0 or 1 and $p(y)$ is the probability that the output belongs to label y . Dice loss is utilized to assess the similarity between the segmentation results from the model and the true labels, as delineated in Eq. (2). By highlighting the overlap between predicted and actual lesion areas, Dice loss effectively counters the negative impact of foreground-background imbalance in samples. This approach significantly mitigates class imbalance in medical imaging datasets, enhancing the model's applicability and accuracy in diverse clinical scenarios.

$$Loss_{dice} = 1 - \frac{2 * (pred \cap true)}{pred \cup true} \quad (2)$$

where $pred$ is the set of predicted values and $true$ is the set of true values, the numerator is the intersection between $pred$ and $true$, and the denominator is multiplied by 2 due to the reason that there exists a double counting of common elements between $pred$ and $true$. the denominator is the concatenated set of $pred$ and $true$. The loss between the predicted value p and the target value t is calculated as shown in Eq. (3).

$$Loss = 0.5Loss_{bce}(p, t) + Loss_{dice}(p, t) \quad (3)$$

In the SC-Unext model's training, the methodology involved the augmentation of datasets through random 90-degree rotations and flipping of images. This augmentation strategy serves to effectively increase the dataset's diversity, enabling the model to recognize and learn from image features presented in various orientations. Such a

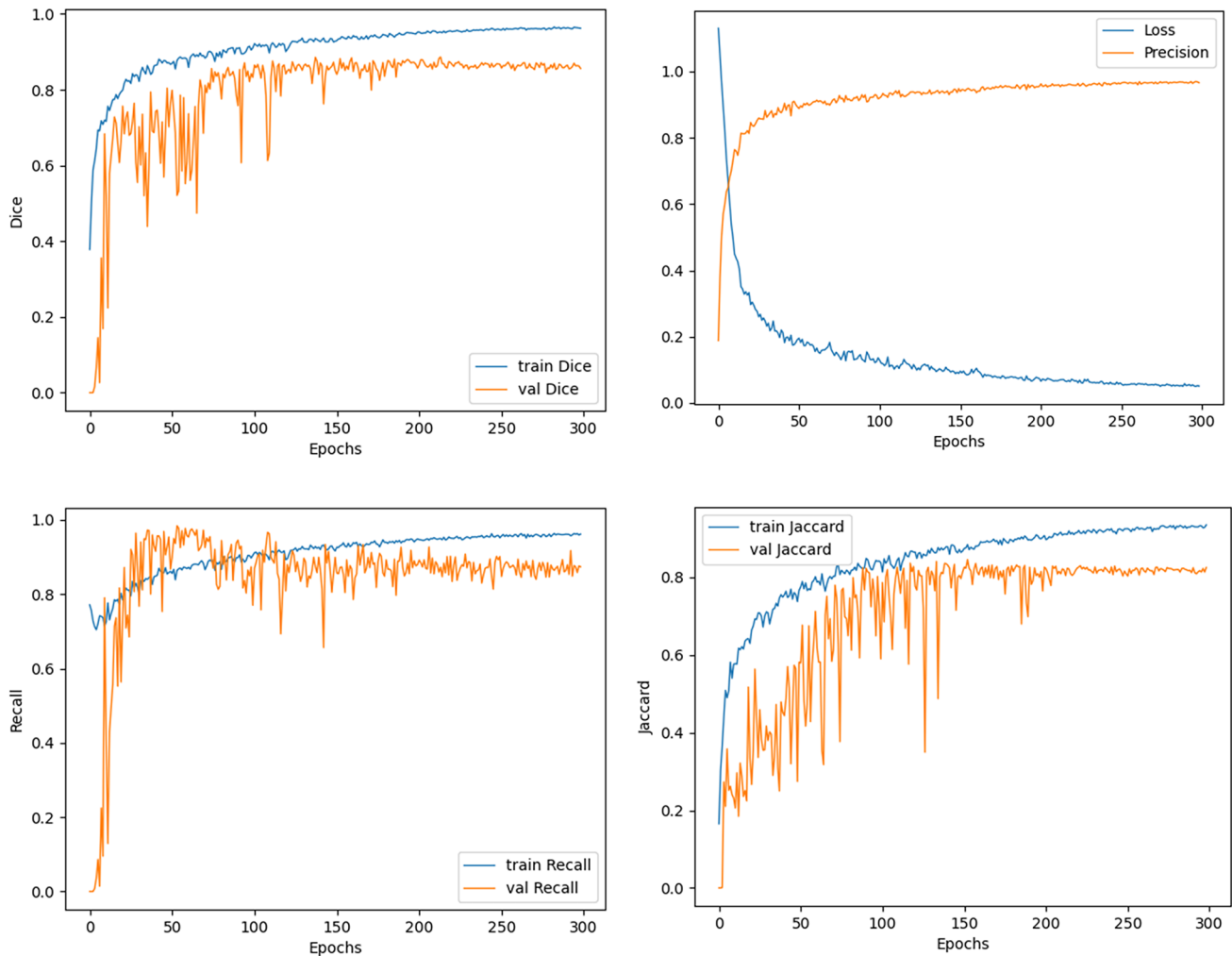


Fig. 3 Performance of metrics in the training and validation process on the CyBUS dataset

Table 1 Comparison of different methods on the BUSI dataset

Method	Dice	Jaccard	Recall	Precision	Accuracy
UNet	70.47±0.21	57.66±0.20	66.12±0.32	80.08±0.42	95.33±0.45
SegNet	71.77±0.21	60.20±0.20	72.03±0.32	79.50±0.42	95.56±0.45
TransUNet	68.94±0.21	57.79±0.43	69.70±0.32	76.28±0.36	93.97±0.23
Unext	72.97±0.31	61.15±0.25	69.96±0.42	80.21±0.45	95.29±0.38
SC-Unext	75.29±0.43	63.66±0.22	75.23±0.35	81.83±0.36	97.09±0.48

technique is instrumental in significantly enhancing the model's generalization capabilities, crucial for robust performance in diverse medical imaging scenarios.

Meanwhile, the learning rate was updated using the cosine annealing learning rate decay strategy, and the Adam optimizer was used to optimize the model training by setting the learning rate to $1e-4$ and the batchsize to 16, and the model was trained for 300 epochs on the two BUSI datasets with the collected breast ultrasound private dataset respectively. Five 80-20 random splits were performed on the BUSI dataset versus the collected breast ultrasound dataset, and the mean and variance of the experimental results are reported. Five widely used segmentation metrics were used in this study to quantitatively compare the different methods, including Dice coefficient [32], Jaccard coefficient, Recall, Precision, and Accuracy [33].

Results

We conducted a series of experiments categorized into different groups, including doctors, Unet, SegNet [34], TransUNet [35], Unext, and SC-Unext (our proposed model). These groups were compared in terms of their segmentation accuracy, computational resource consumption, and segmentation time. The objective was to evaluate the performance of these models in both normal conditions and on resource-constrained devices.

Segmentation with Different Models

In this study, we conducted performance validation of our model on both the BUSI dataset and a privately collected breast ultrasound dataset, as summarized in Table 1. Our enhanced model was compared to the Unext model, resulting

in notable improvements across key metrics. Specifically, we observed a 2.32% increase in Dice, a 2.51% increase in Jaccard, a 5.27% increase in Recall, a 1.62% increase in Precision, and a 1.80% increase in Accuracy.

Furthermore, we assessed the model's performance on the collected breast ultrasound dataset, and the results are presented in Table 2. The model exhibited enhancements of 1.99% in Dice, 3.27% in Jaccard, 2.86% in Recall, 1.82% in Precision, and 0.52% in Accuracy. To provide context, it's worth noting that these improvements contribute significantly to the effectiveness of our model.

Additionally, when compared to other image segmentation networks, our proposed model consistently outperformed them on both the BUSI dataset and the CYBUS dataset.

Figure 3 presents curves illustrating the performance of metrics in the training and validation process on the CyBUS dataset. Notably, during the early stages of training, there is substantial fluctuation in the testing metrics. However, with the increase in the number of training epochs, these evaluation metrics gradually stabilize and reach a plateau, indicating diminishing returns in further improvement.

In Fig. 4, the performance on both the BUSI dataset and the privately collected breast ultrasound dataset is depicted using bar graphs. Subfigure (a) provides a performance comparison among various models on the BUSI dataset, while subfigure (b) offers a comparison of performance on the privately collected breast ultrasound tumor dataset (CYBUS).

Comparison with Doctors

To assess the efficiency of the model for segmenting breast ultrasound images, the manual group involved the participation of two professional ultrasound doctors who manually segmented randomly selected breast ultrasound images. In parallel, the SC-Unext group performed segmentations on

Table 2 Comparison of different methods on the CYBUS dataset

Method	Dice	Jaccard	Recall	Precision	Accuracy
UNet	82.88±0.26	72.54±0.55	80.40±0.50	87.49±0.44	97.37±0.21
SegNet	76.48±0.26	70.12±0.55	82.10±0.50	84.56±0.44	96.91±0.21
TransUNet	83.32±0.16	73.29±0.38	85.58±0.30	83.36±0.24	97.45±0.22
Unext	88.63±0.31	79.80±0.54	88.28±0.57	89.28±0.52	97.85±0.41
SC-Unext	90.62±0.30	83.07±0.70	90.14±0.64	91.10±0.56	98.37±0.37

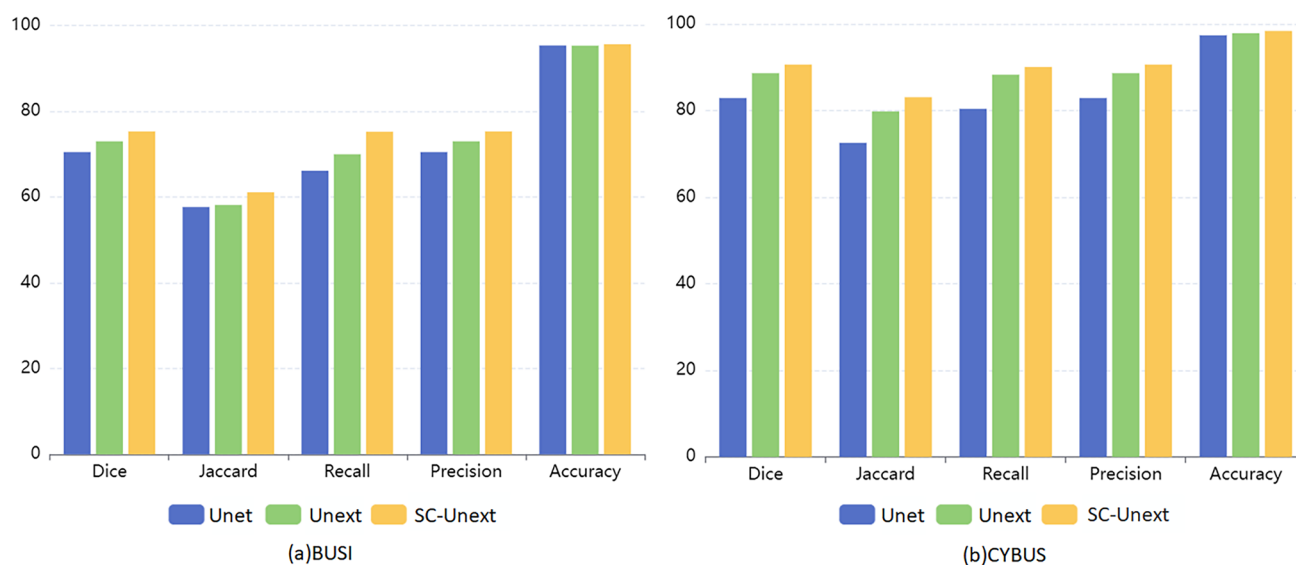


Fig. 4 Comparison of model performance on the BUSI dataset versus the private breast ultrasound dataset

the same set of images. Subsequently, the segmentation results from the doctor group were compared with those from the SC-Unext group to validate the efficiency and accuracy of the SC-Unext model.

As presented in Table 3, Our model demonstrated higher efficiency compared to other methods when processing 60 breast ultrasound images and faster processing speed except Unext. Why SC-Unext is slower than Unext's inference speed is explained in the “[Comparison with Other Models](#)” section. However, recognizing that the computational capabilities of most clinical medical devices are limited without GPU resources, we conducted a comparison of inference speed and segmentation efficiency among the models under resource-constrained conditions, as elaborated in the following section.

Comparison with Other Models

We conducted a comparative analysis of our optimized model with other models, focusing on parameters, computational resource utilization, and inference times, as summarized in Table 4. The SC-Unext model stands out

for its lower parameter count and reduced computational demands compared to other models. While it offers faster inference speeds compared to most models, it's slightly slower than the baseline Unext model. This difference in inference speed can be attributed to the multi-branch structure of the SCConv block, which introduces some computational overhead during inference, impacting the speed relative to Unext.

Figure 5 showcases segmentation results using different methods on two breast ultrasound image datasets. The improved model, after incorporating apoptosis and division algorithms, mitigates some of the challenges related to local optima [36, 37].

Comparison of the Model with Other Models Under Resource-Constrained Device

In real clinical application scenarios, the size of medical devices is a key concern for healthcare professionals. While ensuring accuracy and reliability, the emphasis in medical device development has always been on lightweight and portability at the application end. This is especially crucial for ultrasound devices. However, the pursuit of lightweight

Table 3 Segmentation time of different methods on the CYBUS dataset. time represents the time to segment a single image of a breast tumor, and time (60) is the time to segment 60 images

	Unet	SegNet	TransUnet	Unext	SC-Unext	Doctors	<i>p</i> -value
Time	0.07±0.01 s	0.06±0.01 s	0.07±0.01 s	0.05±0.01 s	0.05±0.01 s	75.47±6.50 s	<0.01
Time (60)	4.61±0.35 s	3.69±0.23 s	4.15±0.33 s	3.23±0.24 s	3.46±0.22 s	4067.56±40.12 s	<0.01
Acc	97.37±0.21	96.91±0.21	97.45±0.22	97.85±0.41	85.71±0.42	92.27±0.25	0.012

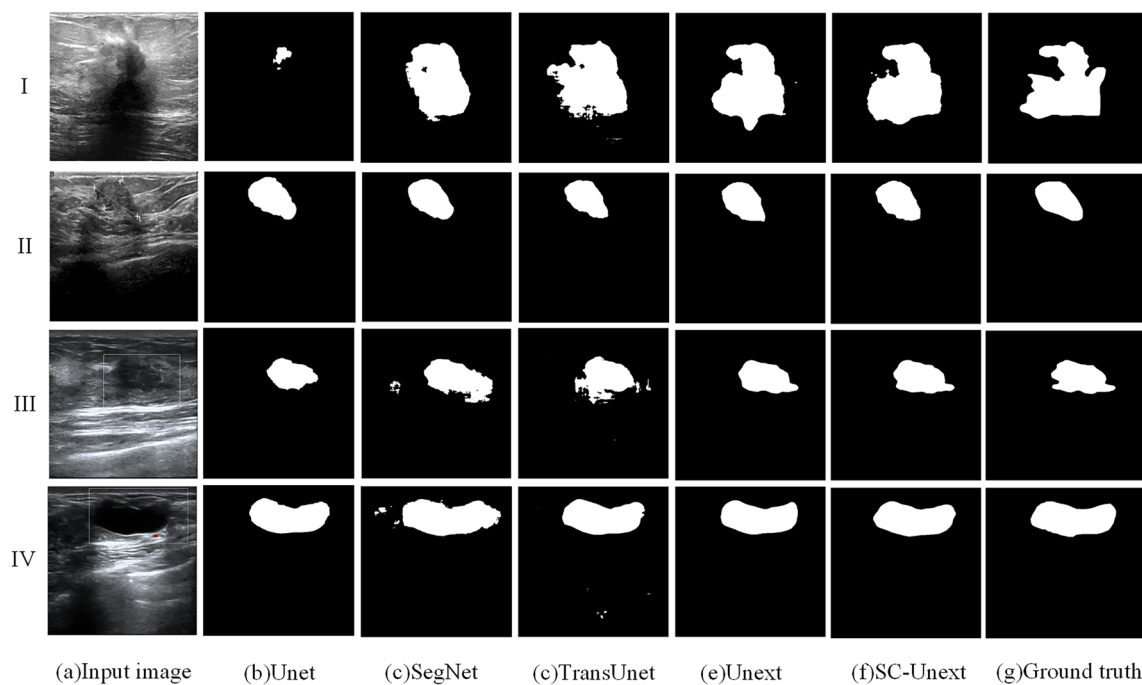


Fig. 5 Segmentation Performance of breast lesions produced by different segmentation methods. I and II are results on the BUSI dataset, III and IV are results on the CYBUS dataset

and even mobile applications (such as tablets or smartphones) inevitably impacts the space available for hosting computational components, resulting in limited computing power at the application end.

Hence, this study conducted a comparison of the model's parameter count, computational resource consumption, and inference speed to investigate its performance in resource-constrained application scenarios in comparison to other models. Given that most clinical devices operate at low computational power without the advantages of GPU computing, this research also conducted comparisons on CPU to assess the model's inference speed under low computational power.

During the inference process, GPU devices operate in different modes under various states. To avoid GPU spending excessive time in the warm-up phase, it is essential to perform pre-warming before inference. The same applies

when using a CPU for inference. Additionally, to ensure the validity of inference time calculations, this study conducted 300 inferences on both GPU and CPU, taking the respective means as the inference speed results.

As indicated in Table 4, the SC-Unext model, with its number of parameters at 1.46 million and computational resource requirement of 2.13 GFLOPs, shows fewer parameter numbers and reduced computational consumption compared to the Unext model. However, the presence of a multi-branch structure in the SCConv module of SC-Unext slightly increases the time cost during parallel computation, resulting in a slightly slower inference speed than Unext. Despite this, SC-Unext maintains a substantial advantage in number of parameters, resource consumption, and inference speed compared to other models. These characteristics make SC-Unext especially suitable for environments with limited or unavailable advanced computational resources, enabling efficient and timely medical image processing on less powerful hardware like standard CPUs without GPU support. This capability is crucial in resource-limited clinical settings for effective and prompt diagnoses.

As shown in Table 5, the SC-Unext model segmented breast ultrasound tumor images on the CPU device is second only to the unext model and significantly faster than the other models, with segmentation times of 14.72 ± 0.14 s and 882.56 ± 21.12 s for a single image and 60 images, respectively. However, the segmentation accuracy of the SC-Unext model is the highest, reaching 0.9837

Table 4 Comparison of resource consumption and inference speed of different methods

	Params	GFLOPs	Inf.T (gpu)	Inf.T (cpu)
Unet	32.08	203.31	26.82	1645.18
SegNet	29.44	160.52	21.41	1645.18
TransUnet	66.81	130.41	25.50	1143.86
Unext	1.47	2.29	4.04	79.94
SC-Unext	1.46	2.13	4.19	92.72

Table 5 Comparison of time to segment single or 60 breast ultrasound tumor images by different methods

Groups	Time (60 photos)	Time (Single)
Unet	1782.56±42.12 s	31.20±0.24 s
SegNet	1782.56±42.12 s	31.20±0.24 s
TransUnet	1782.56±42.12 s	31.20±0.24 s
Unext	20.32±1.64 s	0.34±0.12 s
SC-Unext	24.20±1.44 s	0.42±0.14 s

Discussion

In this study, we introduce the SCConv module to enhance the Unext model and designed algorithms inspired by apoptosis and division mechanisms in multicellular organisms to improve model accuracy and generalization capabilities. Our experiments demonstrate that the improved model outperforms its original counterpart on the BUSI dataset and our collected breast ultrasound dataset.

Multicellular organisms inherently possess significant diversity and adaptability. Mechanisms governing cell division and apoptosis serve as the foundation for adaptive learning and growth within multicellular organisms. Drawing inspiration from the activity mechanisms of multicellular organisms, this research devised algorithms to simulate apoptosis and division, which were then applied to model weight updates. This approach imparts adaptability to the model, enabling it to make appropriate weight adjustments based on data characteristics and variations. Consequently, the model gains the ability to mitigate the challenges posed by local optima to some extent, accelerating the fitting process and enhancing overall performance.

The SC-Unext model is characterized by a reduced number of parameters and lower computational complexity, presenting several advantages over models like Unet. Firstly, the characteristics of fewer parameters and lower computational complexity in the SC-Unext model render it more efficient in terms of memory utilization and processing capability. Secondly, the model's design allows for quick inference, enabling real-time image segmentation. This is crucial for medical imaging applications where speed is essential for timely diagnosis. Furthermore, by incorporating mechanisms inspired by cellular division and apoptosis, and integrating the SCConv module, SC-Unext shows enhanced performance in breast ultrasound image tumor segmentation. In addition, The model's efficiency in computational resource consumption makes it suitable for deployment in environments with limited computational capabilities. Although the SC-Unext model presents numerous benefits, it is not without its drawbacks. The model integrates a multi-branch structure, specifically the SCConv block, which consequently incurs additional computational overhead. This increased complexity marginally diminishes the model's

inference speed when benchmarked against the baseline Unext model. Furthermore, the model's design, characterized by decreased computational resource demands and a fewer number of parameters, might compromise its segmentation efficacy in handling more complex scenarios or tasks.

In the field of medicine, many devices may have limited computational resources. The lower computational resource consumption of the SC-Unext model can reduce computational resource costs for healthcare institutions, making it more adaptable to low-power devices, whether in hardware or cloud-based computing environments. The model's reduced parameter count also translates to lower storage space requirements, which is particularly beneficial for healthcare devices and systems with limited storage resources. Furthermore, SC-Unext can perform inference on local devices, reducing the need to transmit sensitive medical data to external servers and enhancing data privacy protection.

Additionally, compared to manual recognition, SC-Unext can automatically extract regions of interest or lesion areas from medical images without the need for human intervention. This significantly saves on human resources and time while improving segmentation efficiency. The model boasts high accuracy and, in contrast to manual recognition, can precisely capture subtle structural and lesion features, reducing the impact of subjective factors on results. Therefore, the model's rapid inference capability swiftly and accurately segments images, aiding medical professionals in obtaining image segmentation results within a short timeframe. This, in turn, assists doctors in identifying lesion areas more effectively, thereby providing crucial support for clinical decision-making. This support has the potential to enhance diagnostic and treatment efficiency, reduce the risk of misdiagnosis, and is particularly advantageous for clinical applications such as emergencies, rescues, or surgical procedures.

Although the SC-Unext model demonstrates high segmentation precision and lightweight features, it is essential to recognize the limitations inherent in this study. Initially, the research is grounded in an encoder-decoder architecture, refining the Unext network model while preserving its core attributes of light weight and swift segmentation. However, due to reduced computational resource consumption and a decrease in model parameters, the segmentation performance might not be as effective for more complex cases or tasks. Furthermore, the SCConv block, with its multi-branch structure, necessitates a certain amount of time for computational synchronization between branches during model inference, thus constraining the inference speed to some extent, resulting in a performance that is somewhat diminished in comparison to Unext. In addition, the performance and efficiency of SC-Unext model are intimately associated with specific algorithms and frameworks, such as the convolutional neural network encoder-decoder architecture used for medical image segmentation

and the algorithms inspired by the cellular mechanisms of division and apoptosis. This dependency might limit the adaptability of the model to different imaging modalities or new technological advancements in the field.

Conclusions

In the present work, inspired by the mechanisms of biological cells, we introduce a novel model named SC-Unext, which incorporates a Spatial Channel Convolution module and is complemented by algorithms simulating cellular division and apoptosis. We validate the SC-Unext model against other models on the BUSI dataset and a collected dataset, where SC-Unext achieves an Accuracy and Dice coefficient of 97.09% and 75.29% on the BUSI dataset, respectively. On the CYBUS dataset, the model attains an Accuracy and Dice coefficient of 90.62% and 98.37%, respectively. Moreover, compared to other models, SC-Unext requires fewer model parameters and less computational resource consumption. Therefore, SC-Unext holds significant value in the miniaturization and portability of clinical devices.

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Author Contribution All authors contributed to the study conception and design. All authors read and approved the final manuscript.

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Data Availability Due to the nature of this research, participants of this study did not agree for their data to be shared publicly, so supporting data is not available.

Declarations

Ethics Approval This study is designed based on the guidelines for human studies approved by the Ethics Committee of the First Affiliated Hospital of Chongqing Medical University and was ethically in accordance with the Helsinki Declaration.

Informed Consent Informed consent is obtained from all study participants for this study. This research obtains written consent for publication from the parents or legal guardians of all participants in the research.

Conflict of Interest The authors declare no competing interests.

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